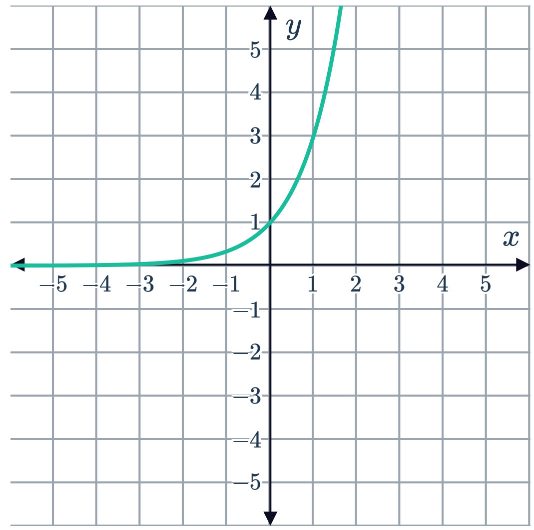
Transformations of exponential functi

1 The graph of $y = 2^x$ is translated down by 7 units, state its new equation.

2 Consider the given graph of $y = 3^x$.



a Describe a transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.

b Sketch the graph of $y = 3^x - 4$.

3 Consider the function $y = 9^x + 5$.

a What value is 9^x always greater than?

b Now, what value is $9^x + 5$ always greater than?

c How many x-intercepts does $y = 9^x + 5$ have?

d State the equation of the asymptote of the curve $y = 9^x + 5$.

4 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

a Find the y-intercept of $y = 2^x$.

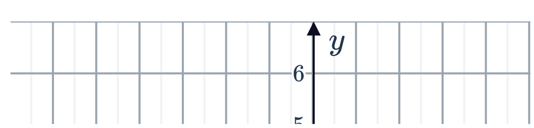
b Now, determine the y-intercept of $y = 2^x - 2$.

c State the horizontal asymptote of $y = 2^x$.

d Now, determine the horizontal asymptote of $y = 2^x - 2$.

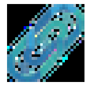
5 Sarah graphs the function $f(x) = 2^x + 6$. She then translated the graph down 9 units to create the function $g(x)$.
State the equation of $g(x)$.

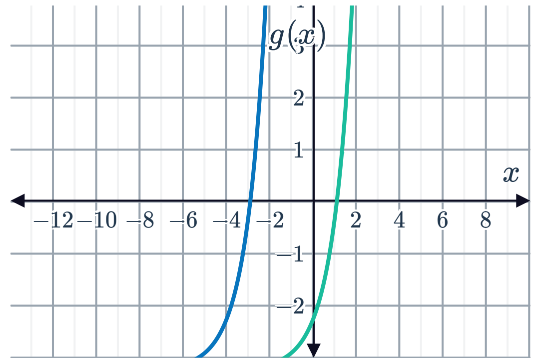
6 Consider the given graphs of $f(x) = 3^x$ and $g(x)$:



a Describe a transformation that can be used to obtain $g(x)$ from $f(x)$.

b State the equation of $g(x)$.

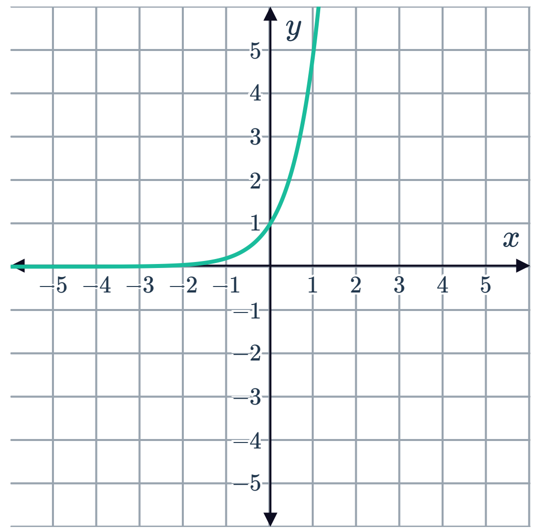




7 Consider the given graph of $y = 5^x$:

a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$.

b Sketch the graph of $y = -5^x$.

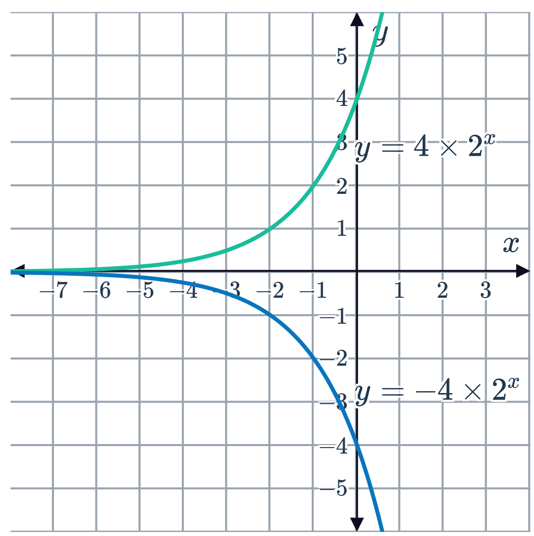


8 Consider the given graphs of $y = 4 \times 2^x$ and $y = -4 \times 2^x$:

a Find the value of y in $y = 4 \times 2^x$ when $x = 0$.

b Describe a transformation of the graph of $y = 4 \times 2^x$ that would obtain $y = -4 \times 2^x$.

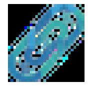
c The number of bacteria over time is to be modeled by an exponential function, with x representing time and y representing the number of bacteria. If the bacteria are increasing, which of the two models should be used?



9 Of the two functions $y = 2^x$ and $y = 3 \times 2^x$, which is increasing more rapidly for $x > 0$?

10 Consider the function $y = 4(2^x)$

a Find the y-intercept of the curve.



b Is the function value ever negative?

c As x approaches infinity, what value does y approach?

d Sketch the graph of $y = 4\left(2^x\right)$.

11 Consider the original graph $y = 3^x$. The function values of the graph are multiplied by 2 to form a new graph.

a For each point on the original graph, find the point on the new graph:

Point on original graph	$\left(-1, \frac{1}{3}\right)$	$(0, 1)$	$(1, 3)$	$(2, 9)$
Point on new graph	$(-1, \square)$	$(0, \square)$	$(1, \square)$	$(2, \square)$

b State the equation of the new graph.

c Sketch the graph of $y = 3^x$ and $y = 2 \times 3^x$ on the same coordinate plane.

d For negative x-values, is the graph of $y = 2 \times 3^x$ above or below $y = 3^x$?

e For positive x-values, is the graph of $y = 2 \times 3^x$ above or below 3^x?

12 Consider the function $y = -2.5 \times 4^x$.

a Is $y = -2.5 \times 4^x$ an increasing or decreasing function?

b As x approaches $-\infty$, what value does y approach?

c As x approaches ∞, y what value does y approach?

d State the y-intercept of the graph.